

Lab 5 - OpenClaw Skills

Extend your agent with pre-built skills. Skills add specialised capabilities - weather, diagrams, browser automation, and more - without writing any code.

Lab environment: Hostinger VPS (from Lab 1) or Docker Desktop (Windows 10/11, macOS 12+, Ubuntu 22.04+)

Prerequisite: Lab 4 completed - agent responding via Telegram or WhatsApp with Ollama model working.

Estimated time: 20 minutes

Docker Desktop users: Prefix every openclaw command with `docker exec -it openclaw`.

Example: `docker exec -it openclaw openclaw skills list`

Step 1 - List Available Skills

```
# VPS
openclaw skills list

# Docker Desktop
docker exec -it openclaw openclaw skills list
```

Skills have two statuses:

- ☐ ready - works immediately, no extra setup needed
- ☐ needs setup - requires a CLI tool or API key installed first

Confirm that weather, diagram-maker, and browser-automation all show ☐ ready before continuing.

Step 2 - Test the Weather Skill

In your Telegram or WhatsApp chat, send:

```
what is the weather in Singapore today?
```

Pass: Agent replies with current temperature and conditions. Fail: "Something went wrong" → model (Ollama) is not running. Run `docker restart openclaw` and retry.

Step 3 - Test the Diagram Maker Skill

In your chat, send:

```
Draw a diagram showing how Docker containers work
```

Pass: Agent replies with an SVG or HTML diagram showing Docker concepts. Fail: Agent replies with text only → send the same message again with "as a diagram" appended.

Step 4 - Test the Browser Automation Skill

In your chat, send:

```
Go to https://news.ycombinator.com and tell me the title of the first story
```

Pass: Agent fetches the page and returns the story title. Fail: Agent says it cannot browse → confirm browser plugin is loaded: `docker exec -it openclaw openclaw plugins inspect browser`

Step 5 - Check Skill Status

```
# VPS
openclaw skills check

# Docker Desktop
docker exec -it openclaw openclaw skills check
```

Pass: All ☐ ready skills show no missing requirements. Fail: A skill shows missing dependency → that skill needs additional CLI setup (not required for this lab).

Step 6 - Install a Skill from ClawHub

Skills not yet installed can be added by name:

```
# VPS
openclaw skills install clawhub:healthcheck

# Docker Desktop
docker exec -it openclaw openclaw skills install clawhub:healthcheck
```

Expected output: `` Installing healthcheck... ☐ Skill installed. ``

Then verify it is ready:

```
# VPS
openclaw skills check

# Docker Desktop
docker exec -it openclaw openclaw skills check
```

Step 7 - Test the Healthcheck Skill

In your chat, send:

```
Run a security healthcheck on this OpenClaw instance
```

Pass: Agent returns a security audit report with pass/fail items.

Step 8 - Update All Skills

```
# VPS
openclaw skills update

# Docker Desktop
docker exec -it openclaw openclaw skills update
```

Updates all installed skills to the latest version.

Ready Skills Reference

Skills marked ☐ ready require no extra setup:

Skill	What it does	Test message
weather	Current weather and forecasts	What is the weather in Singapore today?
diagram-maker	Generate SVG/HTML diagrams	Draw a diagram showing how Docker works
browser-automation	Multi-step browser control	Go to https://news.ycombinator.com and get the first story title
clawhub	Search and install new skills	Search ClawHub for skills related to email
healthcheck	Audit OpenClaw host security	Run a security healthcheck
notion	Read and write Notion pages	List my Notion pages (requires Notion access)
spike	Run throwaway prototypes	Prototype a Python script that sorts a list

Verification

Check	Expected
openclaw skills list	weather, diagram-maker, browser-automation show ☐ ready
openclaw skills check	No missing requirements for ready skills
Chat: What is the weather in Singapore today?	Agent returns current weather
Chat: Draw a diagram showing how Docker works	Agent returns a diagram
Chat: Go to https://news.ycombinator.com and get the first story title	Agent returns story title
openclaw skills install clawhub:healthcheck	☐ Skill installed

Troubleshooting

Symptom	Fix
Skill shows ☐ ready but agent does not respond	Restart: docker restart openclaw then retry
Skill install returns 404	Use exact name from openclaw skills list - do not guess

Symptom	Fix
needs setup skill not working	That skill requires a CLI tool or API key - check openclaw skills check for details
Weather skill returns no data	Try a different city name or check internet connectivity
Diagram not displayed in Telegram	Telegram may not render SVG - ask the agent to "show as an image" instead

Reference

- ClawHub marketplace: <https://clawhub.ai>
- Skills docs: <https://docs.openclaw.ai/skills>